Online processing of morphosyntactic cues by native and non-native Korean speakers: An eye-tracking study

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L1 processing is typically characterized by rapid integration of multiple linguistic cues, whereas L2 processing often relies on a narrower set of salient cues, increasing susceptibility to ambiguity and processing cost (e.g., Bates et al., 1982; Erdocia et al., 2014). Native Korean speakers use case particles and verbal morphology to comprehend sentences with flexible word order to maintain stable performance across structures (Kim, 1999; Lee, 2019). By contrast, Mandarin relies on relatively rigid word order and lacks overt case morphology, potentially leading Mandarin-speaking learners of Korean to underuse Korean morphosyntactic cues and over-rely on word order. This study compares native Korean speakers and Mandarin-speaking L2 learners as they process Korean sentences that involve various structural cues. We hypothesize that learners will show lower accuracy and slower parsing for structures where morphosyntactic cues play a more crucial role in comprehension than surface word order.
Methods (data collection ongoing). 29 native Korean speakers (NS; 23 females; age M=28, SD=4) and 12 intermediate/advanced Mandarin-speaking learners of Korean (NNS; all females; age M=27, SD=3) participated. We used webcam-based eye-tracking with picture selection. On each trial, participants saw two pictures and heard a Korean sentence (Experiment 1: transitive; Experiment 2: dative; Experiment 3: passive), then clicked the matching picture with word order of the sentence manipulated (canonical vs. scrambled) (Examples in 1-3). Accuracy was analyzed with logistic mixed-effects models, response times for correct responses with linear mixed-effects models, and eye gazes with generalized additive mixed models, with Group (NS vs NNS) and Word order as fixed effects and Participant and Item as random effects.
Results. The NS group showed stable accuracy across word orders in all experiments but increased RTs in Experiment 2 (dative) and Experiment 3 (passive) relative to Experiment 1 (transitive) (ps < 0.001). The NNS group was overall slower and less accurate, especially when word order cues conflict with case marking: for scrambled word order conditions, they showed significantly lower accuracy in Experiment 1 (β = -5.180, p < 0.001) and increased RTs in Experiment 2 (β = 0.081, p = 0.002) compared to canonical word order. In Experiment 3, the NNS group's accuracy was overall lower than the NS group. Eye-gaze measures also indicated group differences in Experiments 2 and 3: the NNS looked at the target less than the NS when the second noun was being presented in canonical order in both datives (2055-3388 ms) and passives (2333-3388 ms), and after the second noun (2944-3777 ms) in scrambled passives.
Discussion. Our preliminary findings suggest that native speakers deploy cue-based strategies that integrate case marking, verb morphology, and word order during comprehension, maintaining relatively stable processing across structural manipulations. Relative to native speakers, L2 learners relied more on default word-order expectations. When word-order cues were unreliable (as in scrambled sentences and passives) and morphosyntactic cues had to be prioritized, L2 learners showed processing slowdowns and misinterpretations. With a complete dataset, we expect the results to support cue-based accounts of sentence processing, whereby differences in cue utilization between native and non-native speakers explain L2 processing delays and errors and underscore the importance of attaining more holistic cue integration for efficient Korean sentence comprehension.

	NP1	NP2	Verb
(1) Transitive: ‘Blue hugs Green.’			
a. Canonical	phalangi-ka	choloki-lul	an-ayo.
	blue-NOM	green-ACC	hug-PL
b. Scrambled	choloki-lul	phalangi-ka	an-ayo.
	green-ACC	blue-NOM	hug-PL
(2) Dative: ‘Blue gives Green (a book).’			
a. Canonical	phalangi-ka	choloki-hanthey	cwu-eyo.
	blue-NOM	green-DAT	give-PL
b. Scrambled	choloki-hanthey	phalangi-ka	cwu-eyo.
	green-DAT	Blue-NOM	give-PL
(3) Passive: ‘Green is hugged by Blue.’			
a. Canonical	choloki-ka	phalangi-hanthey	an-ki-eyo.
	green-NOM	blue-DAT	hug-PSV-PL
b. Scrambled	phalangi-hanthey	choloki-ka	an-ki-ess-ta.
	blue-DAT	green-NOM	hug-PSV-PL

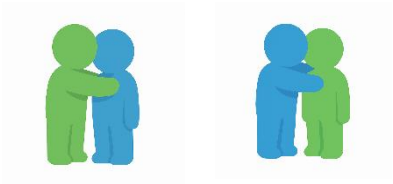

Figure 1. An example trial for the target sentence “Blue hugs Green”. Animacy for both subject and object was controlled via color coding to ensure semantic reversibility and prevent animacy-based heuristics.

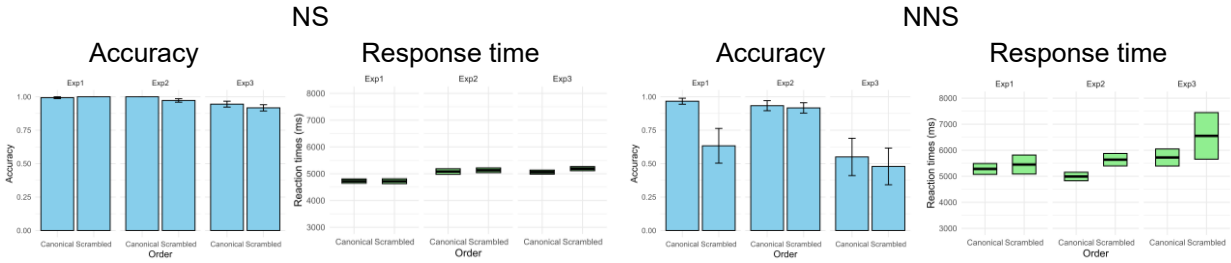

Figure 2. Results from the picture selection task. Error bars indicate standard errors.

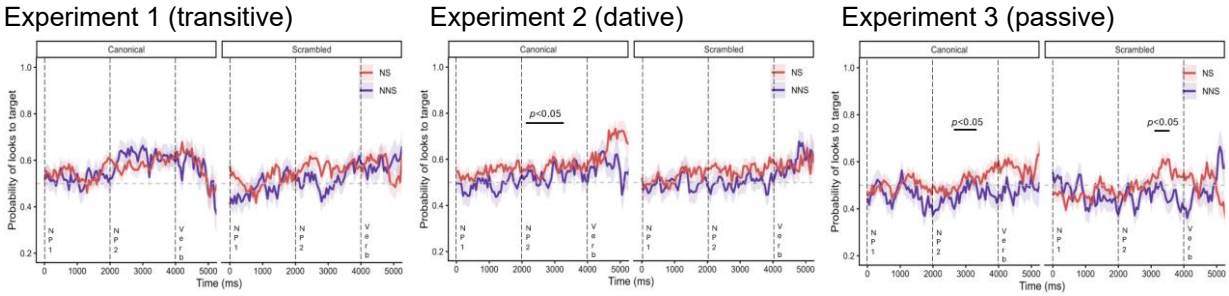

Figure 3. Eye gazes to the target image. Red = NS, blue = NNS; vertical lines = onset of NP1, NP2, Verb in each sentence (duration of each word = approx. 800 ms); black bars = intervals showing statistically significant group differences; shaded areas = standard errors.

Abbreviations. ACC = accusative case marker; DAT = dative marker; NOM = nominative case marker; PL = predicate ending, politeness; PSV = passive suffix.

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